

LDG-A100 RS485 MODBUS 통신

레지스터 주소

레이스터 주소 (기능 코드 04)

Register number	Address	Parameter	Data type	Access Type	Range
3: 0100	0x0063	순간 유량	Float	R	
3: 0102	0x0065	순간유속	Float	R	
3: 0104	0x0067	유량 퍼센트	Float	R	
3: 0106	0x0069	전도도	Float	R	
3: 0108	0x006B	정방향 적산유량	uint32	R	
3: 0110	0x006D	정방향 적산유량 자릿수	uint32	R	The decimal part magnifies by 100 times, 123 stands for 0.123
3: 0112	0x006F	역방향 적산유량	uint32	R	
3: 0114	0x0071	역방향 적산유량 자릿수	uint32	R	The decimal part magnifies by 100 times, 123 stands for 0.123
...	...	Reserve	/	/	Reserve, do not operate
3: 1001	0x03E8	순간유량	Float	R	0~Maximum
3: 1003	0x03EA	정방향 적산유량	Double	R	0~99999999

유량계 통신 레지스터 정보

Function code : 04
해석 순서 3-4-1-2

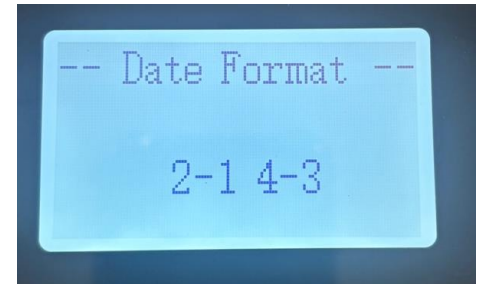
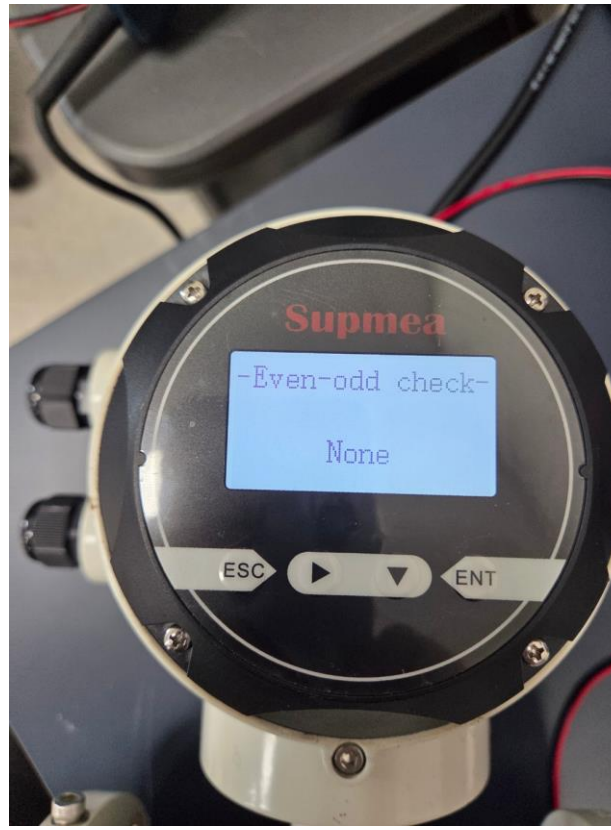
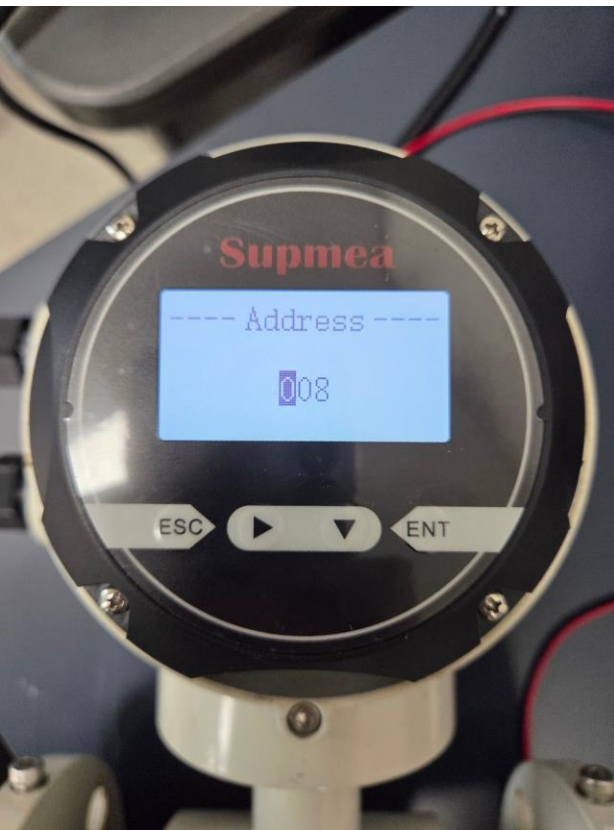
0100~0106 > 32bit float 해석
0108~0114 > 32bit unit 해석

적산 유량의 경우
Unit 32로 해석 시 정수 / 소수 나누어 해석

적산 정수+소수 한번에 볼 경우
레지스터 1003 > IEEE754 Double 해석 필요
해석 순서 3-4-1-2

(현재 저희가 보유중인 테스트 프로그램으로는 Double 해석 불가능하여 32bit 해석 기준으로 안내 드립니다.)

LDG-A100 셋팅



어드레스 : 08
통신 속도 : 9600
패리티 비트 : None parity

해석 순서 3-4-1-2
(제품 2-1 4-3세팅 시 3-4-1-2 순서로 해석)

셋팅 접속 방법

1. 유량계 전면부 set 버튼 눌러 패스워드 1000 입력 후 다시 set 버튼 누르기
2. 3*05~08 통신 설정

LDG-A100 RS485 MODBUS 통신



Simply Modbus Master 8.0.4

mode: RTU, COM port: 7, baud: 9600, data bits: 8, stop bits: 1, parity: None

slave ID: 8, First Register: 40100, No. of Regs: 2

function code: 4, minus offset: 40001, Use defaults: checked, register size: 32 bit registers

Request: 08 04 00 63 00 02 81 4C, SEND

Response: 08 04 04 D1 70 3D B8 4A 81

High byte first: checked, High word first: checked, expected response bytes: 13

SAVE CFG, RESTORE CFG, WRITE, ABOUT

Log: 2025/11/06 16:01:09 >>> 08 04 00 63 00 02 81 4C, 2025/11/06 16:01:11 < 08 04 04 D1 70 3D B8 4A 81

register #: 40100, bytes: D170, results: -6.448923E+10

Analysis Toolpak results table:

		most significant		Least significant	
		high word		low word	
2 words	decimal	16457		4059	
	hex	4049		0FDB	
4 bytes	decimal	64	73	15	219
	hex	3d	b8	d1	80
32 bits	binary	00111101	10111000	11010001	10000000

first bit: 0, invert: 1, SIGN

next 8 bits: 01111011, decimal: 123, subtract 127: -4, EXPONENT

final 23 bits: 01110001101000110000000

if no exponent bits: decimal 3723648, 80 00 00 hex 8388608, divide above and add 1 1.4438934

if no exponent bits: decimal 3723648, 4194304 40 00 00 hex 0.8877869, divide above 1.4438934 MANTISSA (SIGNIFICAND)

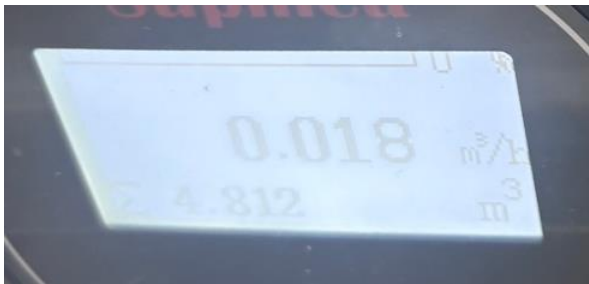
FLOAT = SIGN x MANTISSA x (2 ^ EXPONENT)

FLOAT: 9.0243340E-02

순간 유량 : 0.097 m3/h
Master : 08 04 00 63 00 02 81 4C
Slave : 08 04 **D1 70 3D B8** 4A 81

= 0.0924 m3/h

LDG-A100 RS485 MODBUS 통신



순간 유량 : 0.018 m3/h
Master : 08 04 00 63 00 02 81 4C
Slave : 08 04 **A1 64 3C 98** 4A 81

Simply Modbus Master 8.0.4

mode: RTU, COM port: 7, baud: 9600, data bits: 8, stop bits: 1, parity: None

Slave ID: 8, First Register: 40100, No. of Regs: 2

function code: 4, minus offset: 40001, Use defaults: checked, register size: 32 bit registers

Request: 08 04 00 63 00 02 81 4C

Response: 08 04 04 **A1 64 3C 98** 10 0D

High byte first: checked, High word first: checked, expected response bytes: 13

2025/11/06 16:03:54 >>> 08 04 00 63 00 02 81 4C
2025/11/06 16:03:56 < 08 04 04 A1 64 3C 98 10 0D

This sheet requires the Analysis Toolpak to be loaded. Select the Tools Menu > Add-Ins... > check Analysis Toolpak

		most significant		Least significant	
		high word		low word	
2 words	decimal	16457		4059	
	hex	4049		0FDB	
		high byte		low byte	
4 bytes	decimal	64	72	15	210
	hex	3c	98	a1	64
32 bits	binary	00111100	10011000	10100001	01100100

first bit: 0
invert: 1 SIGN

next 8 bits: 01111001
decimal: 121
subtract 127: -6 EXPONENT

final 23 bits: 00110001010000101100100
if no exponent bits: 1614180
decimal: 80 00 00 hex: 8388608 4194304 40 00 00 hex
divide above and add 1: 1.1924253 0.3848505 divide above
1.1924253 MANTISSA (SIGNIFICAND)

Float = SIGN x MANTISSA x (2 ^ EXPONENT)
Float: 1.8631645E-02

= 0.01863 m3/h